VENKAT ARUN

Assistant Professor, Computer Science Department, UT Austin

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EDUCATION

Massachusetts Institute of Technology (MIT) *2019-2023*
Ph.D. Dept. of EECS
Advisors: Hari Balakrishnan and Mohammad Alizadeh

Massachusetts Institute of Technology (MIT) *2017-2019*
Master of Science, Dept. of EECS
Advisors: Hari Balakrishnan and Mohammad Alizadeh

Indian Institute of Technology Guwahati (IIT-G) *2013-2017*
B.Tech. in Computer Science & Engineering
President of India Gold Medal

INDUSTRY IMPACT

1. Meta uses my congestion control algorithm (CCA), Copa [**??**], for live video uploads
2. Meta uses a my modification to BBR [**??**] (a CCA designed by Google) for a vast majority of their user-facing traffic

HONORS AND AWARDS

- MIT EECS G. M. Sprowls PhD Thesis Award in Computer Science (2024)
- ACM SIGCOMM Doctoral Dissertation Award Runner-Up (2024)
- Marconi Society Young Scholar Award (2023)
- ACM SIGCOMM best student paper award (2022)
- ACM SIGCOMM best paper award (2017)
- MIT Jacobs Presidential Fellowship (2017)
- President of India Gold Medal - IIT Guwahati (2017)
- KVPY Government of India Scholarship (2013)

PUBLICATIONS

- *On the Role of LLMs in Designing the Architecture of Networked Systems*
  Pratyush Sahu, Rahul Bothra, Venkat Arun, Brighten Godfrey, Akshay Narayan, Ahmed Saeed
  NSDI 2027

- *LFS: Learning-Based Flexible CPU Scheduler*
  Yeonju Ro, Vincent Zhang, Venkat Arun, Daehyeok Kim, Jiaxin Lin, Christopher J. Rossbach, Atlas Wang, Aditya Akella
  NSDI 2027

- *Synthesizing a Provably Performant Joint Controller for Low-Latency Video Streaming*
  Jia (Tony) Pan, Anup Agarwal, Isil Dillig, Venkat Arun
  NSDI 2026

- *FRCC: Towards Provably Fair and Robust Congestion Control*
  Anup Agarwal, Venkat Arun, Srini Seshan
  NSDI 2026

- *Contracts: A unified lens on congestion control robustness, fairness, congestion, and generality*
  Anup Agarwal, Venkat Arun, Srinivasan Seshan
  NINeS 2026

- *TierSizer: Counterfactual Reasoning for DRAM Sizing in Tiered Memory Systems*
  Sujay Yadalam, Saarth Deshpande, Divyanshu Saxena, Venkat Arun, Neeraja Yadwadkar, Aditya Akella
  DIMES 2026

- *Bring Your Own Objective: Inter-operability of Network Objectives in Datacenters*
  Sanjoli Narang, Anup Agarwal, Venkat Arun, Manya Ghobadi
  arXiv 2026

- *Faster-than-light coordination for networked systems with quantum non-local games*
  Venkat Arun, Vijay Chidambaram, Scott Aaronson
  HotNets 2025

- *Robust Heuristic Algorithm Design with LLMs*
  Pantea Karimi, Dany Rouhana, Pooria Namyar, Siva Kesava Reddy Kakarla, Venkat Arun, Behnaz Arzani
  arXiv 2025

- *Speculative Ad-hoc Querying*
  Haoyu Li, Srikanth Kandula, Maria Angels de Luis Balaguer, Aditya Akella, Venkat Arun
  arXiv 2025

- *Lightweight Automated Reasoning for Network Architectures*
  Rahul Bothra, Venkat Arun, Brighten Godfrey, Akshay Narayan, Ahmed Saeed
  HotNets 2024

- *Towards provably performant congestion control*
  Anup Agarwal, Venkat Arun, Devdeep Ray, Ruben Martins, and Srinivasan Seshan
  NSDI 2024

- *Hairpin: Rethinking packet loss recovery in edge-based interactive video streaming*
  Zili Meng, Xiao Kong, Jing Chen, Bo Wang, Mingwei Xu, Rui Han, Honghao Liu, Venkat Arun, Hongxin Hu, and Xue Wei
  NSDI 2024

- *A Performance Verification Methodology for Resource Allocation Heuristics*
  Saksham Goel, Benjamin Mikek, Jehad Aly, Venkat Arun, Ahmed Saeed, and Aditya Akella
  arXiv 2024

- *Starvation in end-to-end congestion control*
  Venkat Arun, Mohammad Alizadeh, and Hari Balakrishnan
  SIGCOMM 2023
  Best student paper award

- *Enabling High Quality Real-Time Communications with Adaptive Frame-Rate*
  Zili Meng, Tingfeng Wang, Yixin Shen, Bo Wang, Mingwei Xu, Rui Han, Honghao Liu, Venkat Arun, Hongxin Hu, and Xue Wei
  NSDI 2023

- *Automating network heuristic design and analysis*
  Anup Agarwal, Venkat Arun, Devdeep Ray, Ruben Martins, and Srinivasan Seshan
  HotNets 2022

- *Privid: Practical, Privacy-Preserving Video Analytics Queries*
  Frank Cangialosi, Neil Agarwal, Venkat Arun, Srinivas Narayana, Anand Sarwate, and Ravi Netravali
  NSDI 2022

- *The case for an internet primitive for fault localization*
  William Sussman, Emily Marx, Venkat Arun, Akshay Narayan, Mohammad Alizadeh, Hari Balakrishnan, Aurojit Panda, and Scott Shenker
  HotNets 2022

- *Toward formally verifying congestion control behavior*
  Venkat Arun, Mina T. Arashloo, Ahmed Saeed, Mohammad Alizadeh, and Hari Balakrishnan
  SIGCOMM 2021
  Proposed a modification to BBR that Meta now uses for most of their production traffic

- *Throughput-fairness tradeoffs in mobility platforms*
  Arjun Balasingam, Karthik Gopalakrishnan, Radhika Mittal, Venkat Arun, Ahmed Saeed, Mohammad Alizadeh, Hamsa Balakrishnan, and Hari Balakrishnan
  MobiSys 2021

- *RFocus: Beamforming Using Thousands of Passive Antennas*
  Venkat Arun and Hari Balakrishnan
  NSDI 2020
  May be the largest antenna array ever used for a single communication link

- *Finding safety in numbers with secure allegation escrows*
  Venkat Arun, Aniket Kate, Deepak Garg, Peter Druschel, and Bobby Bhattacharjee
  NDSS 2020

- *Copa: Practical Delay-Based Congestion Control for the Internet*
  Venkat Arun and Hari Balakrishnan
  NSDI 2018
  Being used in production at Meta

- *Language-directed hardware design for network performance monitoring*
  Srinivas Narayana, Anirudh Sivaraman, Vikram Nathan, Prateesh Goyal, Venkat Arun, Mohammad Alizadeh, Vimalkumar Jeyakumar, and Changhoon Kim
  SIGCOMM 2017
  Best paper award

SELECTED PRESS COVERAGE

Starvation in CC	MIT News, IEEE Spectrum, APNIC Blog, The Register, Extreme Tech
RFocus	MIT News, BBC Radio, Tech Crunch, Venture Beat, Engadget, Tech Spot, Digital Trends
Privid	MIT News, IEEE Spectrum, Hacker News, Sci Tech Daily, MarkTechPost